GSoC 2026 Project Proposal

Physics-Informed Shape Optimization of Blade Paul Traps

Organization: ML4SCI

MENTORS

Dale Julson, Eric Reinhardt, Dinesh Ramakrishnan

AUTHOR

Spahr Archibald

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1. Contributor Information

1.1. Personal Information

Name: Spahr Archibald
University / Institution: Université Paris Saclay
Program / Degree: Double Bachelor's degree in Mathematics and Physics
Email: archibald.spahr@gmail.com
Timezone: French Standard Time Zone (UTC +2)

1.2. Academic Background

I am currently pursuing a double Bachelor's degree in Mathematics and Physics at Paris-Saclay University, where I have developed a strong foundation in both theoretical and applied aspects of scientific computing. My coursework includes analysis (real, complex, and functional), linear and abstract algebra, probability, as well as core physics subjects such as electromagnetism, quantum mechanics, and fluid mechanics.

These studies have given me a solid understanding of partial differential equations and physical modeling, which are central to this project. In particular, my background in electromagnetism and numerical methods provides a strong basis for understanding and solving Laplace-type problems in electrostatics.

In September 2026, I will be joining the MSc in Applied Mathematics (Scientific Computing and Machine Learning) at Imperial College London, where I aim to further specialize in scientific machine learning and computational physics. This project is therefore directly aligned with my academic trajectory.

1.3. Past Experience

I have experience programming in Python, PyTorch, SQL and LaTeX, and have worked on projects combining machine learning and mathematical modeling. In particular, during a research internship at the LISV Laboratory (Systems Engineering Laboratory), I implemented a transformer-based model from scratch to generate jazz music, and produced a detailed 40-page technical report explaining the methodology and results (see report).

This experience strengthened my ability to implement complex models, work with modern machine learning frameworks, and understand architectures beyond standard black-box usage. It also developed my skills in structuring experiments, debugging numerical systems, and communicating technical results clearly.

In addition, my academic training has involved extensive use of mathematical tools relevant to this project, including differential equations, linear algebra, and optimization. While I have not previously worked directly on physics-informed neural networks or ion trap design, I have the necessary background and a strong motivation to quickly acquire the required knowledge and implement the proposed framework.

I am also comfortable working independently and managing long-term projects, as demonstrated through both academic work and extracurricular initiatives.

1.4. Availability

I am fully available for the duration of the GSoC program and will be able to dedicate the approximated 175 hours to the project. In case of unavoidable circumstances leading to a temporary reduction in working hours, I will ensure to compensate for the time in the following weeks. I will communicate any such situations with my mentors well in advance to ensure smooth project progress. For regular updates and discussions, I will be available via Teams, Zoom, and GoogleMeet in the French Standard Time Zone (UTC +2), and the Greek one (UTC +4) in august since I will be going to see my family.

I do not have any major conflicting commitments during this period, and I am committed to completing a fully functional, well-documented, and reproducible implementation of the proposed PIG+CPN framework.

2. Abstract

Trapped-ion platforms remain one of the most mature and promising architectures for quantum technologies, including quantum computation, simulation and networking. Among the broad family of radio-frequency ion traps, blade Paul traps are especially attractive because they offer high optical access, and experimentally proven versatility for both single-ion and multi-ion operation. However blade Paul traps have weaker confinement than rod traps.

This proposal addresses that gap by introducing a fully differentiable framework for the *physics-informed shape optimization of blade Paul traps*. The central idea is to couple:

- an adaptive physics-informed solver based on **Physics-Informed Gaussians (PIGs)** for the electrostatic and pseudopotential fields, and
- a **Coordinate Projection Network (CPN)** for continuous shape deformation and optimization.

The motivation for this combination is strong. Classical PINNs are elegant but often suffer from spectral bias and can struggle with sharp gradients, high-frequency behavior, or localized singular structure. PIGs address this by replacing the purely MLP-centered representation with adaptive Gaussian feature embeddings whose means and variances move during training toward physically difficult regions. This makes them particularly well suited to blade traps, where strong field variation appears near electrode edges and tips. In parallel, coordinate projection offers an elegant way to parameterize shape deformation without introducing artificial material smoothing, while keeping geometry differentiable throughout optimization.

The outcome of the project seeks to be an open, modular implementation capable of:

1. solving the electrostatic field of blade Paul trap geometries with a PIG-based physics-informed solver,

2. optimizing blade geometry under trap-quality objectives using a CPN-based shape deformation module,
3. benchmarking against conventional geometries and validating results against numerical baselines.

In short, this project aims to turn trap design from a manual simulation loop into a differentiable optimization pipeline.

3. Paul Traps

3.1. Why Blade Paul Traps?

Blade Paul traps are a class of symmetric radio-frequency traps in which thin electrode blades generate a quadrupole-like field suitable for confining ions. Compared with rod traps, blade traps can offer improved optical access and tighter electrode spacing, which are highly desirable in experiments involving laser access, imaging, and photon collection. They have been used in quantum simulation, precision measurement, and quantum information experiments, and segmented variants allow additional control over axial potentials and ion positioning.

From a design perspective, blade traps are compelling because their performance is highly geometry-dependent. Small changes in blade angle, tip radius, separation, segmentation, and relative alignment can alter the local field curvature, pseudopotential depth, micromotion sensitivity, and optical accessibility. This makes them a natural candidate for differentiable shape optimization.

3.2. Electrostatic Forward Model

We consider the vacuum electrostatic potential $\Phi(\mathbf{x})$ generated by electrodes in a domain $\Omega \subset \mathbb{R}^d$, with $d = 3$. In the vacuum region away from free charge, the governing equation is Laplace's equation:

$$\nabla^2 \Phi(\mathbf{x}) = 0, \qquad \mathbf{x} \in \Omega_{\text{vac}}. \tag{1}$$

Boundary conditions are prescribed on the electrode surfaces:

$$\Phi(\mathbf{x}) = V_{\text{RF}} \cos(\Omega_T t) \quad \text{on RF electrodes,} \tag{2}$$

$$\Phi(\mathbf{x}) = V_{\text{DC},k} \quad \text{on DC or segmented control electrodes,} \tag{3}$$

with appropriate outer-boundary truncation conditions, depending on the domain setup.

Since direct time-dependent simulation of the RF dynamics is typically not needed for geometry optimization, we solve for the *RF amplitude potential*

$$\Phi_{\text{RF}}(\mathbf{x}) \tag{4}$$

such that the time-harmonic physical potential is

$$\Psi(\mathbf{x}, t) = \Phi_{\mathrm{RF}}(\mathbf{x}) \cos(\Omega_T t). \tag{5}$$

3.3. Pseudopotential Approximation

Near the trap center, the time-averaged ponderomotive pseudopotential for an ion of charge q and mass m is

$$U_{\mathrm{ps}}(\mathbf{x}) = \frac{q^2}{4m\Omega_T^2} \|\nabla \Phi_{\mathrm{RF}}(\mathbf{x})\|^2. \tag{6}$$

This quantity defines effective confinement in the radial directions. Good trap geometries are those for which the pseudopotential has:

- a well-defined and centered RF null,
- strong but controlled local curvature,
- acceptable depth and secular frequencies,
- robustness to small perturbations and misalignments.

3.4. Secular Motion and Stability Quantities

In an ideal quadrupole setting, ion motion can be described by the standard Mathieu equations. However, for geometry optimization, it is not necessary to simulate full ion trajectories. Instead, stable confinement can be characterized through field-derived quantities computed from the RF amplitude potential and the associated pseudopotential.

In this work, we focus on metrics that directly correspond to the optimization objectives used in the learning framework:

- **RF null position:** The location where the electric field vanishes, ensuring minimal micromotion and proper centering of the trap.
- **Pseudopotential depth:** The effective trapping depth defined by the pseudopotential barrier, which determines the robustness of confinement against escape.
- **Geometric accessibility constraints:** Measures related to electrode proximity and opening angles, which impact optical access and experimental usability.

These quantities provide a physically meaningful and computationally efficient surrogate for full dynamical simulation, and are directly used to define the optimization losses guiding the geometry design.

4. Technical Approach and Model Design

4.1. Overview of the Proposed Architecture

The proposed system consists of two tightly coupled modules:

1. a **PIG-based field solver** that predicts the RF amplitude potential and associated field quantities in the domain,
2. a **shape optimization module** based on a Coordinate Projection Network, which maps a reference blade geometry into an optimized geometry.

4.2. Physics-Informed Gaussians (PIGs) as the Field Solver

4.2.1. Motivation

PIGs are an attractive solver for this project because they retain the familiar physics-informed optimization framework of PINNs while replacing the rigid MLP-centric representation with adaptive Gaussian feature embeddings. Each Gaussian has a learnable center, scale, and feature weight. During training, Gaussians can migrate toward regions where the solution is more difficult to approximate, such as high-gradient or singular areas.

This is especially relevant for blade traps. In electrostatics, the potential itself may remain smooth in the vacuum region, but the field magnitude and curvature can vary sharply near the electrode boundaries and especially near blade tips or narrow openings. A solver that can reallocate representation capacity spatially is therefore highly desirable.

4.2.2. Field Representation

Let $\mathbf{x} \in \mathbb{R}^d$ denote a coordinate. We represent the solution through a Gaussian feature extractor:

$$\mathrm{FE}_\phi(\mathbf{x}) = \sum_{i=1}^{N} f_i \, G_i(\mathbf{x}), \tag{7}$$

where

$$G_i(\mathbf{x}) = \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_i)^\top \boldsymbol{\Sigma}_i^{-1}(\mathbf{x}-\boldsymbol{\mu}_i)\right), \tag{8}$$

with learnable Gaussian centers $\boldsymbol{\mu}_i$, covariance parameters $\boldsymbol{\Sigma}_i$, and feature vectors f_i.

A lightweight neural refinement head then predicts the potential:

$$\Phi_{\phi,\omega}(\mathbf{x}) = \mathrm{NN}_\omega(\mathrm{FE}_\phi(\mathbf{x})). \tag{9}$$

In practice, the network head can be extremely small compared to a classical PINN, because most of the representational burden is carried by the Gaussian basis layer.

4.2.3. Why PIGs Are Well Matched to Blade Traps

The decision to use PIGs is not arbitrary. Blade Paul traps create a geometry in which:

- relevant features are *localized*,
- some regions require much higher representational capacity than others,
- the field behavior near edges matters disproportionately for downstream trap metrics.

PIGs match this regime well:

1. Gaussian kernels are local, unlike a global MLP basis.
2. Their centers and scales are adaptive, so capacity can migrate toward difficult regions.
3. Gaussian functions are smooth and differentiable, which is ideal for computing PDE residuals and higher-order derivatives.
4. The resulting model remains physics-informed and mesh-free in the training sense, while still behaving like an adaptive parametric mesh.

4.2.4. Physics-Informed Loss for the PIG Solver

The field solver will be trained with a composite loss:

$$\mathcal{L}_{\text{PIG}} = \lambda_{\text{PDE}}\mathcal{L}_{\text{PDE}} + \lambda_{\text{BC}}\mathcal{L}_{\text{BC}}$$

(10)

The PDE residual term is:

$$\mathcal{L}_{\text{PDE}} = \frac{1}{|\mathcal{X}_\Omega|} \sum_{\mathbf{x}\in\mathcal{X}_\Omega} \left|\nabla^2 \Phi_{\phi,\omega}(\mathbf{x})\right|^2, \tag{11}$$

where $\mathcal{X}_\Omega$ are sampled interior collocation points.

The boundary condition term is:

$$\mathcal{L}_{\text{BC}} = \frac{1}{|\mathcal{X}_{\partial\Omega}|} \sum_{\mathbf{x}\in\mathcal{X}_{\partial\Omega}} \left|\Phi_{\phi,\omega}(\mathbf{x}) - V(\mathbf{x})\right|^2. \tag{12}$$

Optional symmetry constraints may be added for trap configurations designed to preserve mirror or rotational structure. Regularization can also be used to control pathological covariance growth, redundant Gaussian overlap, or drift away from the region of interest.

4.2.5. Expected Advantages Over PIXEL-Like Parametric Grids

A natural alternative would be a fixed-grid parametric representation such as PIXEL-style models. Those approaches are promising, but for blade-trap optimization they remain less attractive than PIGs. The main reason is that fixed-grid methods place representational resources at predetermined collocation locations, whereas PIGs can move those resources adaptively. In the present application, this matters because field difficulty is not uniform across the domain. The most challenging structures are precisely the localized regions near edges, gaps, and tip singularities. A static grid wastes parameters in uninformative regions, while adaptive Gaussians can migrate to the locations where additional resolution is needed most.

4.2.6. Sampling Strategy

The success of the PIG solver will also depend on a robust sampling policy.

- uniform interior collocation points for global consistency,
- boundary oversampling on electrode surfaces,
- boundary oversampling near tricky geometric regions
- residual-driven adaptive resampling as training progresses.

This complements the adaptive Gaussian representation and should improve both accuracy and data efficiency.

4.3. Shape Optimization via Coordinate Projection Networks (CPNs)

4.3.1. Why Choose CPNs for Shape Optimization?

The shape optimization module must satisfy three requirements:

1. it must be differentiable,
2. it must support nontrivial geometric deformations,
3. it must remain compatible with PDE training on domains with discontinuous material or boundary descriptions.

Coordinate Projection Networks satisfy these requirements elegantly. Rather than defining the design through a material density field or a smoothed indicator function, a CPN learns a mapping

$$\mathbf{x} = T_\psi(\mathbf{z}), \tag{13}$$

from reference coordinates $\mathbf{z}$ to deformed coordinates $\mathbf{x}$.

4.3.2. CPN Formulation

Let Ω_z denote the reference computational domain and $\Omega_x = T_\psi(\Omega_z)$ the deformed physical domain. The shape network is:

$$T_\psi : \Omega_z \to \Omega_x. \tag{14}$$

If $\mathbf{z}$ is a point in the reference geometry, its deformed physical coordinate is:

$$\mathbf{x} = T_\psi(\mathbf{z}). \tag{15}$$

The Jacobian of the mapping is:

$$J_\psi(\mathbf{z}) = \det \nabla_{\mathbf{z}} T_\psi(\mathbf{z}). \tag{16}$$

To avoid folding and preserve feasible geometry, we enforce:

$$J_\psi(\mathbf{z}) > 0 \tag{17}$$

through penalty terms. This makes sure the transformation stays a diffeomorphism, making sure there is no change in topology of the shape, since CPN's can't handle them.

4.3.3. Shape Loss Terms

The shape network is trained jointly with the field solver through a combination of trap-quality and geometric regularization losses:

$$\mathcal{L}_{\text{shape}} = \lambda_{\text{jac}} \mathcal{L}_{\text{jac}} + \lambda_{\text{vol}} \mathcal{L}_{\text{vol}} \tag{18}$$

Each term plays a distinct role in ensuring that the learned deformation improves trap performance while remaining geometrically valid.

Jacobian Regularization $\mathcal{L}_{\text{jac}}$. This term penalizes invalid local deformations by constraining the determinant of the Jacobian of the shape map T_ψ:

$$\mathcal{L}_{\text{jac}} = \frac{1}{|\mathcal{X}_{\text{shape}}|} \sum_{\mathbf{z} \in \mathcal{X}_{\text{shape}}} \left[\text{ReLU}(J_{\min} - \det \nabla T_\psi(\mathbf{z})) + \text{ReLU}(\det \nabla T_\psi(\mathbf{z}) - J_{\max}) \right]^2. \tag{19}$$

It prevents folding, inversion, or collapse of the geometry.

Smoothness Loss $\mathcal{L}_{\text{smooth}}$. This term penalizes rapidly varying or oscillatory deformations:

$$\mathcal{L}_{\text{smooth}} = \frac{1}{|\mathcal{X}_{\text{shape}}|} \sum_{\mathbf{z} \in \mathcal{X}_{\text{shape}}} \|\nabla^2 T_\psi(\mathbf{z})\|^2. \tag{20}$$

It encourages smooth, fabrication-compatible electrode shapes.

Together, these losses ensure that the shape optimization is driven by trap-quality improvement while remaining stable, physically meaningful, and geometrically realistic.

4.4. Coupling the Shape Module with the PIG Solver

The full optimization loop is:

1. start from a reference blade geometry,
2. deform the geometry using CPN ,
3. sample interior and boundary points in the deformed domain,
4. solve for the RF potential using PIGs,
5. compute pseudopotential and trap-quality objectives,
6. backpropagate through the entire pipeline to update both solver and shape parameters.

The total loss may be written as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{PIG}} + \mathcal{L}_{\text{shape}} + \lambda_{\text{access}}\mathcal{L}_{\text{access}}, \tag{21}$$

where

Access Loss $\mathcal{L}_{\text{access}}$. This term encourages geometries that improve optical or physical access to the trapping region by penalizing obstruction or proximity of electrodes to the trap center. A generic formulation is:

$$\mathcal{L}_{\text{access}} = \frac{1}{|\mathcal{X}_{\text{ROI}}|} \sum_{\mathbf{x}\in\mathcal{X}_{\text{ROI}}} \mathcal{P}(\mathbf{x}), \tag{22}$$

where $\mathcal{P}(\mathbf{x})$ measures geometric obstruction or distance-based penalties relative to the electrodes.

5. Experimental Plan

5.1. Validation Strategy

Validation is performed at three levels:

- **Physics:** PDE residual and boundary-condition accuracy,
- **Numerical:** comparison with external solvers,
- **Design:** geometric plausibility and trap-quality improvement.

5.2. Failure Modes to Monitor

Key risks include:

- insufficient resolution in the PIG solver,
- instability or collapse of Gaussian parameters,
- invalid or folded geometry from the CPN,
- objective-driven but physically unrealistic shapes,

6. Expected Deliverables

By the end of the project, I aim to deliver the following:

6.1. Scientific Deliverables

- an open-source implementation of a PIG-based electrostatic solver for blade Paul trap geometries,
- a differentiable CPN shape optimization module,
- experiment scripts and reproducible training configurations.

6.2. Documentation Deliverables

- developer documentation for the architecture,
- tutorial notebooks for a minimal trap optimization example,
- final report summarizing results, limitations, and next steps.

7. Timeline and Deliverables

7.1. Community Bonding Period

- Finalize project scope and success criteria with mentors.
- Review the relevant codebase, development practices, and contribution workflow.
- Refine the reference blade-trap parameterization and evaluation metrics.
- Set up the software environment, experiment tracking, and documentation structure.

7.2. Weeks 1–2: Geometry and Baseline Setup

Goals:

- Implement a reference blade-trap geometry generator.
- Implement domain and boundary samplers for interior and electrode points.
- Build a baseline classical PINN for Laplace solves.
- Define validation benchmarks against a trusted numerical solver.

Deliverables:

- reference geometry module,
- sampling utilities,
- baseline PINN implementation,
- initial validation results on fixed geometries.

7.3. Weeks 3–4: PIG Solver Implementation

Goals:

- Implement the Gaussian embedding layer and lightweight refinement network.
- Benchmark PIG against the baseline PINN on fixed electrostatic problems.
- Visualize Gaussian movement and residual concentration during training.

Deliverables:

- working PIG electrostatic solver,
- benchmark notebook or experiment report,
- ablation study on Gaussian count and sampling strategy.

7.4. Weeks 5–6: CPN Shape Module Implementation

Goals:

- Implement the Coordinate Projection Network as a near-identity deformation map.
- Add Jacobian and smoothness regularization for stable geometric deformations.
- Define the trap-quality objective based on RF null position and multipole coefficients.

- Validate that the CPN produces smooth and physically meaningful deformations.

Deliverables:

- working CPN deformation module,
- implementation of geometric regularization losses,
- initial trap-objective evaluation pipeline.

7.5. Weeks 7–8: Joint PIG+CPN Optimization

Goals:

- Couple the PIG solver and CPN into a single end-to-end differentiable framework.
- Optimize the geometry using the combined trap-quality and shape-regularization losses.
- Evaluate whether the model can improve RF null positioning and multipole quality.

Midterm Deliverables:

- end-to-end PIG+CPN prototype,
- first optimized blade-trap geometry results,
- midterm report summarizing implementation progress and initial findings.

7.6. Weeks 9–10: Robustness and Extended Evaluation

Goals:

- Improve training stability through better weighting, initialization, and sampling.
- Evaluate optimization performance on multiple reference blade geometries.
- Analyze the effect of the learned deformations on trap-quality metrics

Deliverables:

- stabilized training pipeline,
- comparison across multiple geometries,
- quantitative evaluation of optimization gains.

7.7. Weeks 11–12: Final Experiments and Documentation

Goals:

- Run the final benchmark suite and collect reproducible results.
- Prepare visualizations of optimized geometries, field solutions, and Gaussian adaptation.
- Finalize documentation, tutorials, and usage examples.
- Write the final report summarizing methods, results, and limitations.

Deliverables:

- cleaned and documented code contribution,
- reproducible experiment scripts,
- tutorial examples,
- final report and presentation-ready summary.

7.8. Buffer and Risk Management

A successful GSoC project needs explicit contingency planning. I therefore include the following safeguards:

- If joint optimization of the PIG+CPN system is unstable, training will be staged, first validating the PIG solver independently before introducing geometric deformation.
- If blade Paul traps are too complex for stable shape optimization within the project timeframe, the focus will shift to either:
  - developing and validating a high-quality PIG solver for PT electrostatics, or
  - applying the PIG+CPN shape optimization framework to a simpler geometry where optimization behavior can be more reliably demonstrated.
- If optimization objectives conflict or lead to unstable training, the project will prioritize a single well-motivated primary objective (e.g., multipole quality or RF null positioning) with minimal auxiliary constraints.
- If training becomes sensitive to sampling or hyperparameters, the project will focus on a simpler model.

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